

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. (CE) Examination****April-May 2018****CE202.01/CE 202: DATA COMMUNICATION & NETWORKING****Date: 04/05/2018, Friday****Time: 10:00 am To 01:00 pm****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make & specify suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q – 1 Do as Directed. [05]**

- (a) What is mean by Data Communication? [01]
- (b) What are the advantages of Distributed Processing? [01]
- (c) Why are Standards needed? [01]
- (d) For n devices in a network, what is the number of cable links required for a mesh and ring topology? [02]

Q - 2 Do as Directed (Any Three). [15]

- (a) Discuss these topologies in details along with advantages and disadvantages:
Ring Topology and Star Topology
- (b) Explain Circuit-Switched Connection versus Packet-Switched Connection.
- (c) What is the role of Transport Layer and Presentation Layer in OSI model? Explain services/functionalities of these two layers in detail.
- (d) What are the applications of Computer Network?
- (e) Calculate CRC and find out reminder by considering these data: frame 110101 and generator 101.

Q - 3 Do as Directed (Any Three). [15]

- (a) What do you mean by ARPANET? Explain it in detail.
- (b) Explain five principles that were applied to arrive at the seven layers of OSI layer.
- (c) What are the basic components for working of a communication system?
- (d) What do you mean by Virtual Circuit? What do you mean by SVC and PVC? Explain them in detail.
- (e) What are the key functions of error control techniques?

SECTION – II**Q – 4 Do as Directed [05]**

- (a) What is Multiplexing? [01]
- (b) Define Bandwidth and Latency in Computer Network. [02]
- (c) How does NRZ-L differ from NRZ-I? [02]

Q - 5 Do as Directed (Any Three).

[15]

- (a) Discuss advantages and disadvantages of Fiber Optic for network communication.
- (b) Explain Serial Transmission and Parallel Transmission in detail.
- (c) Draw pattern for 01001100011 according to following two methods:
Bipolar-AMI and Pseudoternary
- (d) Explain Time and Frequency Domain Representation of signals
- (e) Draw and Explain 802.3 frame format.

Q - 6 Do as Directed (Any Three).

[15]

- (a) Explain process of Cyclic Redundancy Check (CRC) both at Sender and Receiver side.
- (b) What is Time-Division Multiplexing? Explain it in detail with example.
- (c) What is Pulse Code Modulation (PCM)? Explain this in detail.
- (d) Compare and Contrast Switch and Hub.
- (e) Explain following propagation modes for Signal:
 - 1. Sky-wave propagation
 - 2. Line-of-sight propagation
